STUDENT SUMMER INTERNSHIP TECHNICAL REPORT

Development of Double Shell Tank Annulus Vacuum Crawler System

DOE-FIU SCIENCE & TECHNOLOGY WORKFORCE DEVELOPMENT PROGRAM

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FIU | Applied Research Center
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EXECUTIVE SUMMARY

This research work has been supported by the DOE-FIU Science & Technology Workforce Development Initiative, an innovative program developed by the U.S. Department of Energy's Office of Environmental Management (DOE-EM) and Florida International University's Applied Research Center (FIU-ARC). The initiative was supported by Sr. Research Scientist Dr. Amer Awwad. During the summer of 2025, DOE Fellow intern Ronald Verde undertook a 10-week internship at Hanford Tank Waste Operations & Closure (H2C) Chief Technology Office (CTO), with guidance from Dr. Douglas J. Reid and Rogeiro G. Renteria. The project, which began on May 28, 2024, and concluded on August 9, 2024, continued on June 9, 2025, and concluded on August 14, 2025, focused on the development of a robotic vacuum crawler aimed at enhancing inspection and cleaning operations within the annulus space of double-shell tanks (DSTs). The work was aligned with the needs of H2C's Tank and Pipeline Integrity (TAPI) organization, as outlined in the Technology Element Description Summary (TEDS) MTW-89. This document highlights the necessity of remote cleaning for concrete surfaces painted with a special protective coating (SPC), required for Resource Conservation and Recovery Act permit compliance, to enable proper inspection and repair in high-radiation environment.

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TRADEMARKS

- *Exair Line Vac 6080 is a registered trademark of EXAIR, 11510 Goldcoast Dr Cincinnati, OH, USA.*
- *Intel® RealSense™ D415 camera is a registered trademark of Intel Corporation, USA.*
- *Stratasys F170 3D printer is a registered trademark of Stratasys, 7665 Commerce Way Eden Prairie, MN USA.*

LIST OF ACRONYMS

ABS	Acrylonitrile butadiene styrene
CTF	Cold Test Facility
CTO	Chief Technology Office
DOE-EM	Department of Energy's Office of Environmental Management
DSTs	Double-shell tanks
FIU	Florida International University
FIU-ARC	Florida International University's Applied Research Center
H2C	Hanford Tank Waste Operations & Closure
PCB	Printed Circuit Board
SCFM	Standard Cubic Feet per Minute
SPC	Special protective coating
TAPI	Tank and Pipeline Integrity
TEDS	Technology Element Description Summary
TLR	Technology Readiness Level
VCS	Vacuum Crawler System

1 INTRODUCTION

During the summer of 2025, Ronald Verde, a mechanical engineering student from Florida International University (FIU), joined the Chief Technology Officer (CTO) at Hanford Tank Waste Operations & Closure (H2C) to contribute to a key research and development project for the Tank and Pipeline Integrity (TAPI) organization. This project, which focuses on remote cleaning solutions for the annulus space of double-shell tanks (DSTs), directly supports the U.S. Department of Energy's environmental remediation goals. This work is executed in alignment with the strategic vision of the Technology and Innovation Roadmap, RPP-PLAN-43988 (now in Rev.09 [1]), which identifies the need for the development of robotic platforms and remote technologies for radiological surveying, inspections, and manipulations to significantly reduce human exposure to hazardous environments. The specific need for this remote cleaning solution is detailed in the Technology Element Description Summary (TEDS) MTW-89, which highlights the requirement for remote cleaning of concrete surfaces painted with a special protective coating (SPC). The core of this project focused on the design and development of the remotely operated robotic vacuum crawler intended for the inspection and cleaning of the annulus space in double-shell tanks (DSTs).

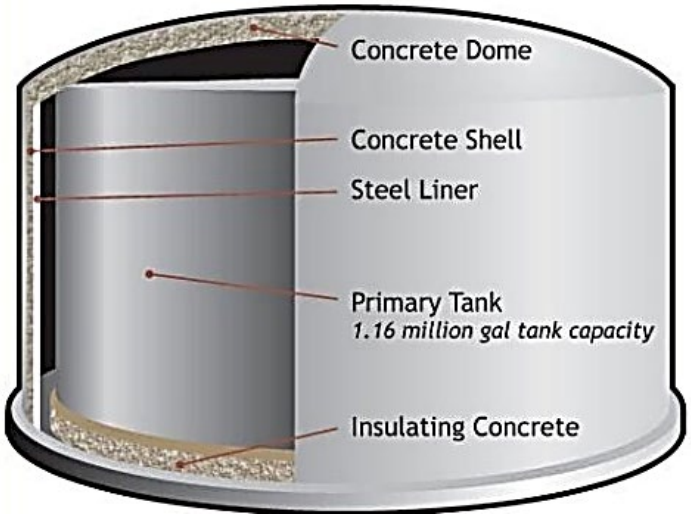


Figure 1. (28) The Underground Double-Shell Tanks constructed in 1968-1986

Hanford's Double-Shell Tanks (DSTs) are critical infrastructure for storing radioactive waste, and their annular spaces must be regularly inspected and maintained to ensure structural integrity, as shown in Figure 1. A primary challenge in this process is the accumulation of various forms of debris which hinder crucial inspection procedures such as ultrasonic testing. The Annulus Vacuum Crawler project was developed to address this issue by providing a robotic solution to navigate these confined and hazardous spaces to remove debris, thereby enabling proper inspections. The urgency of this project has been heightened by the introduction of new debris, specifically powdered residue from recent cold spray repairs, which adds to the original debris on the annulus floor.

1.1 Project Outline

Over time, Hanford's Double Shell Tanks accumulated debris, with some tanks experiencing buildup as deep as 1 foot, as shown in Figure 2. The annulus space approximately 30 inches wide with access points ranging from 12 to 24 inches in diameter, presents significant engineering challenges for robotic systems designed to operate within such confined environments.

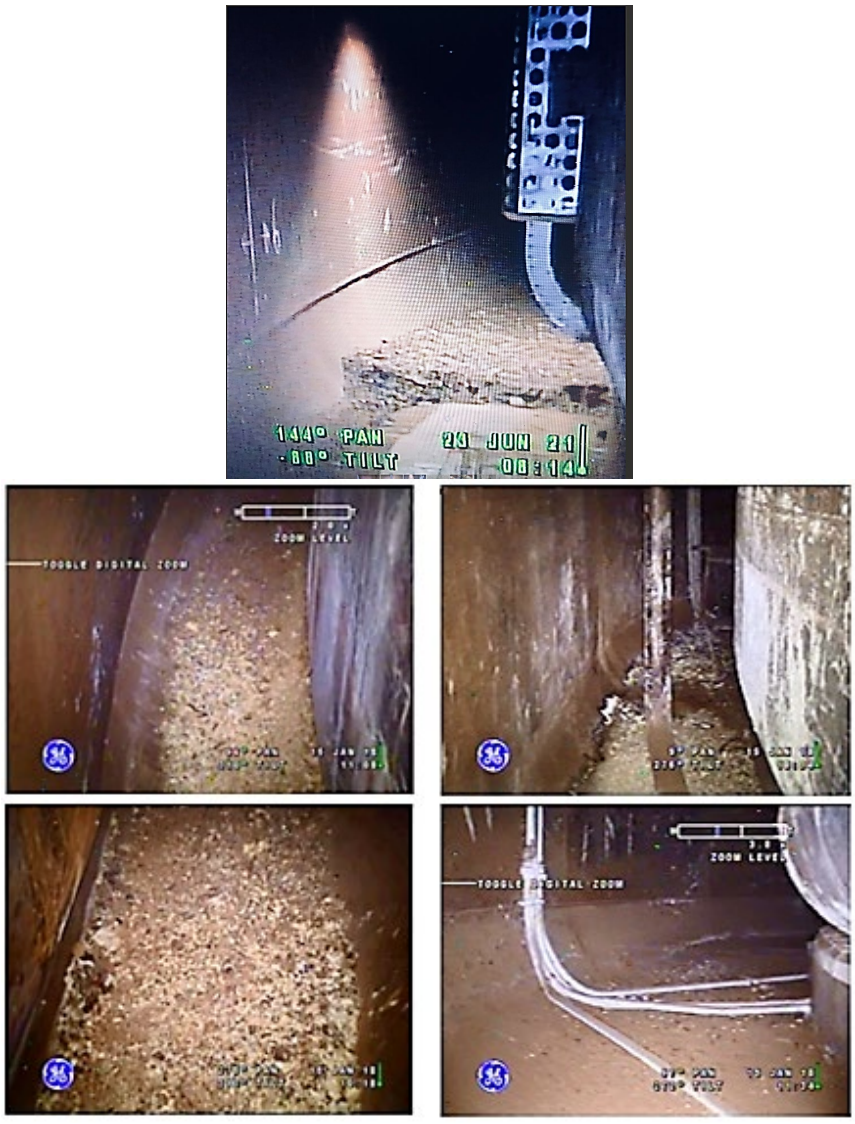


Figure 2. Debris Inside of the Annulus Double Shell Tank AY-101

This report examines the methods, testing processes, and project outcomes associated with annulus waste management. It highlights the hazardous nature of the annulus space and its critical role in maintaining nuclear site integrity. In alignment with the U.S. Energy Sector's Environmental Management Program, continuous development efforts aim to improve the reliability, efficiency and safety of the systems. The scope of the 2025 summer internship will focus on advancing the Annulus Vacuum Robot from a laboratory-validated system to an operational-ready prototype.

These developments are a refinement of a tool that enhances H2C's Tank Inspection capabilities by adding to the operations' toolbox. Improvements on this tool will be directly guided and validated by these groups, who will directly determine the system's readiness for test deployment. These stages of validation would be analogous to DOE's Technology Readiness Levels (TRL) 4 and 6 respectively. The internship work will focus on critical research, development, testing, and refinement activities executed through targeted methodological advancements for the Annulus Vacuum Robot system.

2 RESEARCH DESCRIPTION

2.1 Deployment strategies

The initial stage of the internship focused on developing a deployment strategy for the crawler through both the 24-inch and 12-inch Mockup Risers at the Cold Test Facility, as photographed in Figure 3 . This deployment strategy, depicted in Figure 4, involves using an electric hoist to drop the crawler face down. The design ensures that the crawler's wheels make initial contact with the annulus floor, allowing it to drive forward and orient itself correctly upon landing. Key considerations included managing the descent to ensure the wheels touch down first, thereby preventing the front nozzle from impacting the floor. This method also allows for manual retrieval using a safety suspension cable in the event of a mechanical failure.



Figure 3. Cold Test Facility Double- Shell Annulus Tank Deployment Mock (Left) and the Cold Test Facility (Right)

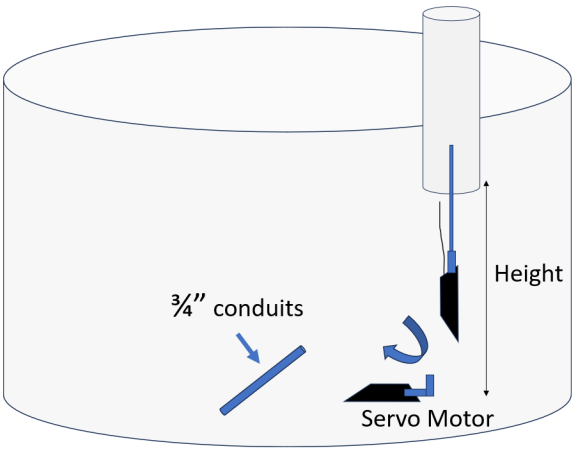


Figure 4. Diagram of the Developed Deployment Strategy

The operators' exceptional skill in handling the hoist system was a major factor in the success of our mock demonstrations. Their ability to make small, precise adjustments during the descent made the process significantly smoother and more controlled. Furthermore, their established control room, seen in Figure 5 , located 500 feet away from the tank, was equipped with an external camera that provided them with enhanced visibility. This remote vantage point was critical, allowing the team to carefully monitor the crawler's deployment and ensure a seamless entry into the annulus.



Figure 5. Established Control Room.

A series of essential modifications were made directly to the crawler, eliminating the need for the external harness that was previously required. A newly manufactured cage and an onboard bag for debris collection were added to the crawler itself. This streamlined the entire system, making the separate, suspended debris collector obsolete. As a result of these improvements, the main connection to the crawler is now limited to just the vacuum hose, electrical cable, level cable, and safety cable. This simplified setup improves deployment and retrieval efficiency, while also minimizing the risk of entanglement within the confined space of the tank annulus.

2.2 Design

The project's primary objective is to develop a tool capable of removing debris from the Double-Shell Tank annulus, thereby facilitating accurate and thorough inspection of the tank walls for thickness and integrity. This work directly supports the Hanford Tank Waste Operations & Closure (H2C) mission.

2.1.1 On-Board Debris Collection Cage

To achieve this, several key design components were developed and refined. The on-board debris collection cage was designed for both strength and functionality. Its frame structure was built using SolidWorks' 3D sketch capabilities, with swept features forming a curved top section to fit the 12-inch riser. This design allows the operator to quickly and easily dispose of collected debris.

2.1.2 Advanced Hub Design

The hub assembly also underwent significant customization. The purchased hubs were too large, prompting the team to design a compact, integrated solution in SolidWorks. This custom design ensures a perfect fit with our specific wheel and motor shaft, providing direct and efficient power transfer to the wheels.

2.1.3 Enhanced Chassis for Vacuum System

The chassis was redesigned to accommodate the new vacuum system and better navigate the tank environment. Integrated channels were added to route airflow and wiring internally, minimizing external hoses and snag hazards. The chassis also allows the crawler system to traverse conduits, and modular service panels created using advanced surfacing techniques to provide easy maintenance access to all internal components.

2.1.4 Redesigned Vacuum Head

The vacuum head was completely redesigned to improve performance. SolidWorks' cavity feature and flow simulation analysis were used to create a multi-zone suction system with parametric control for optimized debris pickup. The design included an adjustable height mechanism with linear actuators and guide rails, allowing adaptation to uneven surfaces. Internal baffles, created with swept cuts and lofted features, prevent oversized objects from clogging the system. The new head is specifically sized to efficiently vacuum debris up to 1/4-inch in diameter and includes an onboard collection bag to reduce travel distance for debris, increasing efficiency.

2.1.5 Crawler System Upgrades

Key upgrades during the internship included the fabrication of a new all-aluminum chassis, replacing the previous 3D-printed plastic frame. This enhancement significantly increased the crawler's durability and radiation tolerance. The system also features newly designed treaded aluminum wheels for improved traction and the ability to cross obstacles up to ½ inch in height. Its compact dimensions allow it to pass through both 12-inch and 24-inch riser access points, and the design enables 180-degree maneuverability within the 30-inch-wide annular space, a critical feature for effective navigation. To further enhance structural integrity and support the new onboard debris collection system, a CNC-machined cage was manufactured to secure the equipment bag. The chassis prototype is illustrated in Figure 6.

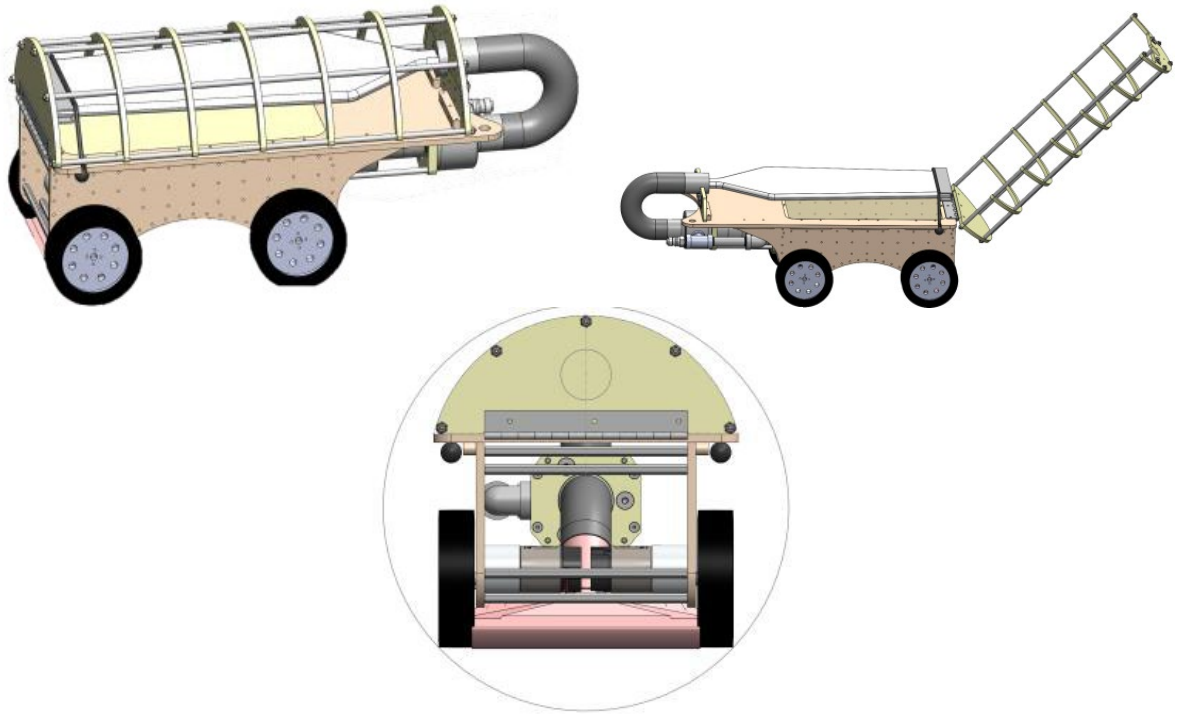


Figure 6. Chassis prototype

2.3 Vacuum System Design & Development

The robot is built around a two-line vacuum configuration using two Exair Line Vac pneumatic vacuum modules positioned in series to optimize suction power. One module is mounted directly on the crawler, while the other is integrated on the top of the crawler itself. This design ensures maximum airflow and pulls force at the cleaning nozzle. During development, the coupling size was changed from 3/8" to 1/2" to increase the volume of suction. For the demonstration, a 1-inch hose line was used, which was then connected to a 1/2-inch hose line leading to the crawler. An appropriately sized portable air compressor as seen in Figure 7 rated at 185 standard cubic feet per minute (SCFM), provides the necessary flow rate.



Figure 7. Air Compressor and Air Lines.

The vacuum crawler system (VCS) is built around powerful vacuum architecture. The previous design configuration, which had one module on the umbilical harness, has been superseded. In the modified model, two EXAIR Line Vac pneumatic vacuum modules are now installed in series and onboard the crawler itself. This was made possible by new alterations to the chassis and the design of a CNC-machined cage that allows the debris collection bag to be mounted directly on the crawler. This system-integrated module maximizes both vacuum force and airflow at the cleaning nozzle. The pneumatic vacuum system is powered by an appropriately sized portable air compressor, rated at 185 SCFM, which provides the necessary flow rate.

2.4 Electronics & Controls

Completely modernized the system's electronics and controls by implementing a new, custom-designed Printed Circuit Board (PCB). This PCB created a highly organized and reliable connection between the main components, including the Raspberry Pi, Teensy, RoboClaw motor controllers, and the motors themselves. This consolidated architecture eliminates the traditional mess of wires and significantly enhances signal integrity and overall system stability.

The new control interface features a custom control box, seen in Figure 8, with an integrated touch screen that runs on a Linux with Ubuntu operating system. This setup provides a professional and intuitive user experience. Replaced the traditional joystick with a Logitech controller for simplified and precise operation. This allows for seamless control of the robot, its cameras, and the on-board lights. The touch screen also provides a platform for displaying live, real-time video feedback, giving the operator crucial situational awareness.

To manage the system's functions, developed specialized ROS2 (Robot Operating System 2) nodes. These nodes are the core of our control logic, handling everything from the motor functions for the wheels to the servo motor that controls the camera's movement. Used "joy nodes" and "telejoy nodes" to translate inputs from the Logitech controller into precise commands for the robot. A separate ROS2 node was created for our dual-camera setup, and a dedicated launch script was developed to configure and activate all these nodes simultaneously. This robust integration allows for real-time navigation and remote control, all supported by a carefully designed power management system that ensures extended operational time.

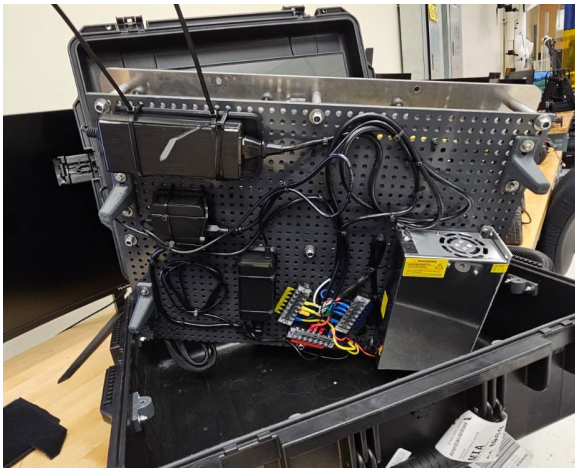


Figure 8. New Control Interface Featuring a Custom Control Box.

3 RESULTS AND ANALYSIS

The VCS underwent rigorous, multi-stage testing at both the FIU Robotics Lab and the Cold Test Facility (CTF) to evaluate its performance under conditions simulating the annulus space of the DSTs. These evaluations focused on the system’s core subsystems: vacuum performance, maneuverability, and control system responsiveness. Deployment strategies were tested in a mock DST environment, seen in Figure 9, and the crawler’s onboard microcontrollers were assessed for reliability and integration, seen in Figure 10.



Figure 9. Deployment in the Mock Double Shell Tank at Cold Test Facility.

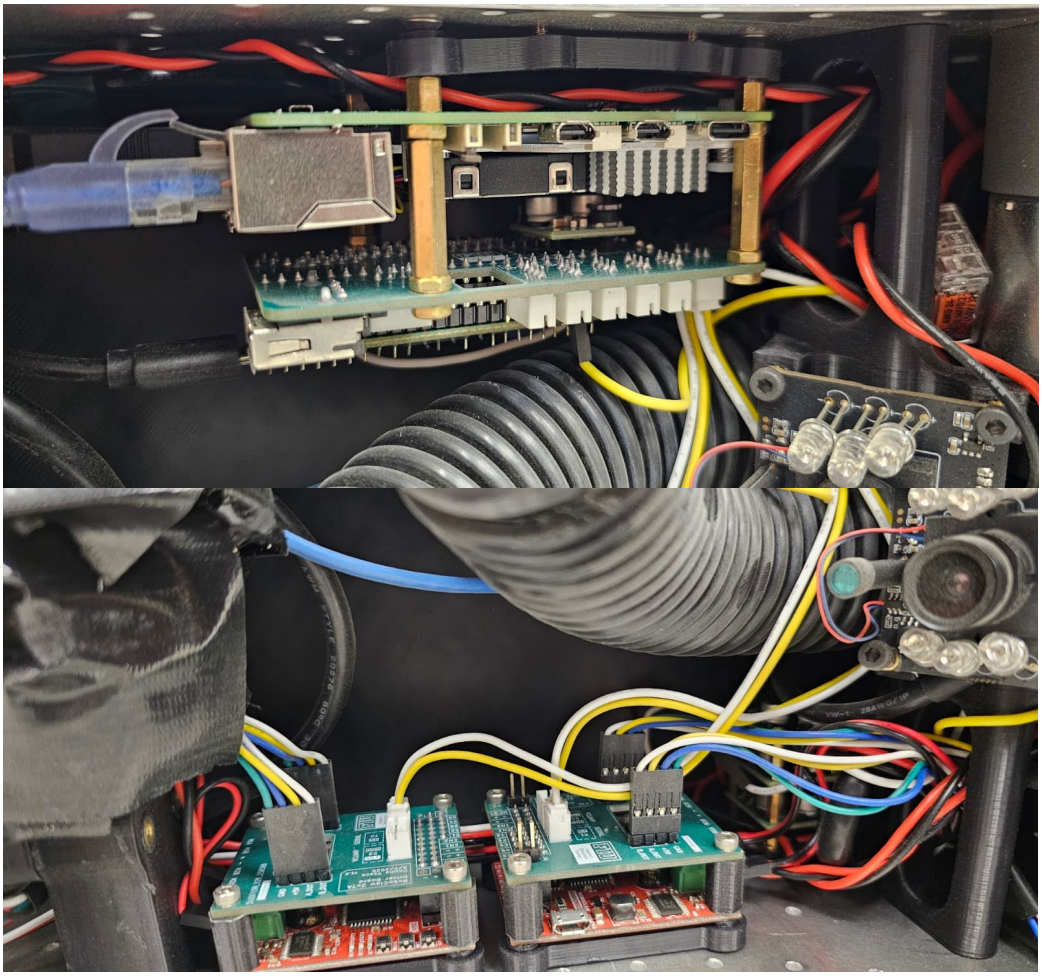


Figure 10. Microcontrollers in Crawler. Performance and Efficiency Testing:

Vaccum Performance Testing involved assessing the suction power and airflow of the onboard two-line Exair Line Vac modules, which were mounted in series on a newly designed aluminum chassis. This configuration, supported by a CNC- machined cage for the onboard debris collection bag, eliminated the need for an external umbilical harness and demonstrated effective debris removal, when tested with various aggregates of debris and cold spray powder. The onboard dual-module setup enhanced airflow and reduced the risk of clogging, demonstrating the viability of the new, more compact design. Efficiency was further validated using a Sullair air compressor, shown in Figure 11. capable of delivering 185 SCFM at 120 psig, ensuring optimal airflow and pressure. The Line Vac Venturi injector design, Figure 12, enabled high output flows while remaining maintenance-free due to the absence of moving parts.



Figure 11. Sullair Air Compressor Used for Validation of the Vacuum Performance.

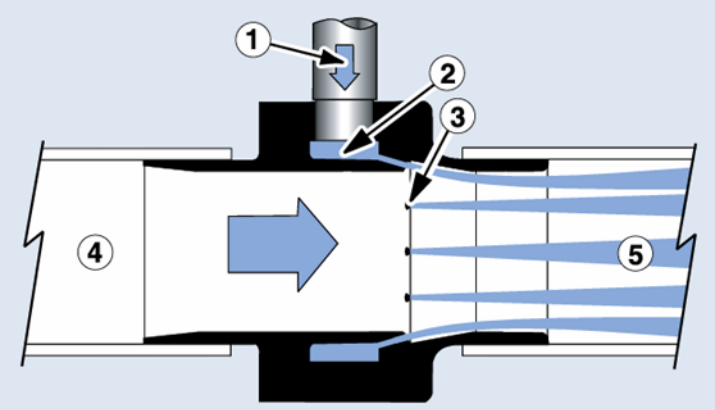


Figure 12. LineVac Venturi Injector Design.

3.1 Maneuverability Testing

The crawler's ability to traverse obstacles and navigate around welds up to ½ inch in height was thoroughly tested. The new aluminum chassis and treaded wheels were evaluated to ensure reliable performance on warped liner bottoms and slope surfaces within the annulus space. The differential drive control system was fine-tuned to optimize the crawler's responsiveness and stability during operation, and the crawler successfully demonstrated 180° maneuverability within the confined 30" wide space.

3.2 Control System Validation and Performance

The VCS's control system, comprising a newly developed control box, custom PCBs, and ROS2-based architecture, underwent thorough testing to evaluate its responsiveness, reliability, and operational precision. The system demonstrated excellent responsiveness to operator inputs, with ROS2 nodes reliably managing camera feeds and teleoperation functions. Real-time video feedback with low latency was validated, ensuring effective remote operation and precise crawler navigation within the confined annulus space. The physical setup of the control box connected to the crawler during testing is shown in Figure 13. illustrating the integration if hardware components used to validate system performance. The Elimination of traditional analog circuits and joystick interfaces marked significant improvement in system simplicity and stability. The advanced controller streamlined operation and enhanced user control, allowing operators to maintain crawler stability during complex maneuvers. Overall, the control system's performance confirms that the current design meets the fundamental requirements for DST annulus cleaning. Insights gained from this comprehensive analysis will guide further refinement of the system, focusing on optimizing vacuum efficiency and ensuring the crawler's durability and reliability in the nuclear environment.

4 CONCLUSION

The development and testing of the vacuum crawler system during this internship successfully demonstrated its capability to address the critical challenges associated with the inspection and cleaning of the annulus space in Double-Shell Tanks (DSTs) at DOE's Hanford site. The crawler's robust new aluminum chassis ensured reliable maneuverability, and the new ROS2-based control system provided precise operation with low-latency video feedback, contributing to the overall effectiveness of the system. The crawler successfully performed its primary function of debris removal, thus validating its design and operational capabilities.

The success of these initial testing trials indicates that the vacuum crawler system (VCS) is well-positioned to advance into further stages of development, with the potential to significantly enhance the safety and efficiency of hazardous waste management in nuclear facilities. Insights gained from these trials will guide the next steps in refining the system's design. Planned improvements for future iterations include:

- Full integration of the sensor suite with obstacle avoidance capabilities.
- Deployment of a comprehensive Graphical User Interface (GUI) for operator monitoring.
- Continued suction efficiency testing with various debris types, including cold spray powder.
- Radiation-hardening of components to achieve full in-tank readiness.
- Achieving Technology Readiness Level (TRL) 6 with a final CTF test and validation.

Additional improvements will focus on addressing:

- Weight distribution by adding ballast to the front of the crawler to achieve a more balanced center of gravity.
- Implementing a fail-safe start/stop system to enhance operational safety.
- Investigating the use of larger wheels to further improve obstacle traversal capabilities.
- Employing a non-moving vacuum head to optimize debris collection efficiency.

This project has demonstrated the viability of using remotely operated robotic systems for complex environmental management tasks, contributing to the broader mission of the U.S. Department of Energy's environmental remediation efforts. With continued development, the VCS has the capability to become a valuable tool in supporting the maintenance of nuclear storage facilities. The System’s successful deployment, as shown in Figure 12, underscores its readiness for further testing and integration into hazardous waste management operations.



Figure 13. Successful Deployment and Control System Operation.

5 REFERENCES

[1] RPP-PLAN-43988 Rev.09, "Technology and Innovation Roadmap, Rev. 9," Washington River Protection Solutions, LLC, Richland, Washington, 2024.

[2] Department of Energy. Accessed on August 20, 2024. Site : https://www.energy.gov/

6 APPENDIX

Technology Element Description Summary (TEDS) MTW-89:

RPP-PLAN-43988
Revision 9

Technology and Innovation Roadmap

Prepared by

A.L.Hede
Washington River Protection Solutions, LLC

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September 2024



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Office of River Protection

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A-6007-231 (REV 0)



RPP-PLAN-43988, Rev. 9

CHIEF TECHNOLOGY OFFICE

washington river protection solutions

Remote Concrete Surface Cleaning Apparatus

HANFORD SITE
US DEPT OF ENERGY

A remotely operated cleaning tool for concrete walls, waste transfer pits, ceiling cover blocks that are required to be painted with a SPC. This device must be able to thoroughly clean the SPC (i.e., Amerlock® 2/400 resin) without damaging it.

TEDS ID: MTW-89 **Timetable: > 5 Years**

TECHNOLOGY NEED

A technology is needed to allow for remote cleaning of concrete walls, waste transfer pits and ceiling cover blocks that are painted with a special protective coating (SPC). Per WAC-173-303, SPC is required for concrete surfaces that may come in contact with tank waste (e.g., tanks vaults process pits, valve pits) must be inspected and repaired as necessary to maintain RCRA permit compliance. These surfaces will get dirty and contaminated due to occasional spills during operations. Before proper inspection and repair can occur the surfaces must be cleaned. Due to high radiation fields noted in most pits, the work associated with the cleaning and required coating repairs must be completed remotely (through the use of extension poles). Inspection of the cleaned and repaired surfaces is completed by visual inspection of high quality digital photographs.

TECHNOLOGY SOLUTION

The DST Annulus Floor Cleaning System, developed by Rolls-Royce for use by WRPS, is a remote operated cleaning tool designed to vacuum debris in DSTs annulus. It is believed that this technology could be further modified to be used in transfer pits and other RCRA facility locations to perform the necessary cleaning.



Typical Transfer Pit Configuration, Areas that Need to be Cleaned



Robotic Cleaner with Vacuum

Technology Maturation Level

Research and Concept

National Laboratory Involvement?

No

Rough Order of Magnitude Cost & Duration?

$0 - $1 Million

0 Years - 2 Years

THREATS AND OPPORTUNITIES

DAOR24-0016-T: Pit Corrosion

DSTI24-0001-T: DST Failure

TLI24-0001-T: Waste Transfer Line Failure While Operating

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Phone: (509) 372-3629
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